Today

- The Schrödinger equation in three dimensions
- Angular momentum
- Orbital angular momentum and spherical harmonics

4 Three-dimensional systems

Now, we begin investigating the Schrödinger equation for a particle in three dimensions with time-independent potential $V(\vec{r})$.

4.1 Three-dimensional infinite square well

Consider a particle of mass m in potential $V(x, y, z) = 0$ for $0 \leq x \leq l_x$, $0 \leq y \leq l_y$, $0 \leq z \leq l_z$, and $V(x, y, z) = \infty$ outside. Because the potential is cubic, the Schrödinger equation in cartesian coordinate is suitable. Therefore, the time-independent Schrödinger equation becomes

$$-\frac{\hbar^2}{2m}\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}+\frac{\partial^2}{\partial z^2}\right)\psi(x,y,z) = E\psi(x,y,z) \tag{4.1.1}$$

for $0 \leq x \leq l_x$, $0 \leq y \leq l_y$, and $0 \leq z \leq l_z$, while $\psi(x, y, z) = 0$ outside. Because there is no potential that mixes the different coordinates, we can use the method of separation of variables, and start with writing,

$$\psi(x, y, t) = \psi_1(x)\psi_2(y)\psi_3(z). \tag{4.1.2}$$

The Schrödinger equation then becomes

$$-\frac{\hbar^2}{2m}\frac{d^2}{dx^2}\psi_1(x) = E_1\psi_1(x), \quad -\frac{\hbar^2}{2m}\frac{d^2}{dy^2}\psi_2(y) = E_2\psi_2(y), \quad -\frac{\hbar^2}{2m}\frac{d^2}{dz^2}\psi_3(z) = E_3\psi_3(z), \tag{4.1.3}$$

where

$$E = E_1 + E_2 + E_3. \tag{4.1.4}$$

We know the solutions from the one-dimensional infinite square well,

$$\psi_{n_x n_y n_z}(x, y, t) = \sqrt{\frac{8}{l_x l_y l_z}}\sin(\frac{n_x\pi}{l_x}x)\sin(\frac{n_y\pi}{l_y}z)\sin(\frac{n_z\pi}{l_z}z) \tag{4.1.5}$$

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and

$$E_{n_x n_y n_z} = \frac{\hbar^2 \pi^2}{2m}\left(\frac{n_x^2}{l_x^2} + \frac{n_y^2}{l_y^2} + \frac{n_z^2}{l_z^2}\right) \tag{4.1.6}$$

with $n_x, n_y, n_z = 1, 2, ...$ Note that the wave vector is

$$\vec{k} = (k_x, k_y, k_z) = (\frac{n_x \pi}{l_x}, \frac{n_y \pi}{l_y}, \frac{n_z \pi}{l_z}). \tag{4.1.7}$$

Exercise 4.1.1: When we talked about Planck's black-body radiation formula, we mentioned that the number density (per unit volume per unit frequency) of states in a cavity was $\frac{\mathcal{D}(\omega)}{V} = \frac{\omega^2}{\pi^2 c^3}$. Show that considering the standing modes of the three-dimensional infinite square well in a large cubic box.

Solution: Considering a large box, we can view its volume in $\vec{k}$–space as spherical, and the volume of spherical shell of infinitesimal thickness dk at radius k is $\frac{1}{8} 4\pi k^2 dk = \frac{1}{2}\pi k^2 dk$, where the factor of $\frac{1}{8}$ because all components of $\vec{k}$ are positive. The volume of a single cell is $\vec{k}$–space is $\frac{\pi}{l_x}\frac{\pi}{l_y}\frac{\pi}{l_z} = \frac{\pi^3}{V}$, where $V = l_x l_y l_z$ is the volume of the box. The number of states within this volume is $dN = 2\frac{\frac{1}{2}\pi k^2 dk}{\frac{\pi^3}{V}} = \frac{V}{\pi^2} k^2 dk$, where the factor of 2 is because radiation has two possible polarizations, or a standing plane wave has two independent directions (for describing the plane on which the electric field lives). Considering radiation, $k = \frac{\omega}{c}$. Thus, $dN = 2\frac{V}{\pi^2}\frac{\omega^2}{c^2}\frac{d\omega}{c}$ and $\frac{\mathcal{D}(\omega)}{V} = \frac{dN}{d\omega V} = \frac{\omega^2}{\pi^2 c^3}$.

Exercise 4.1.2: The Hamiltonian operator for a particle of mass m constrained to undergo circular motion on a frictionless track is given by $\hat{H} = -\frac{\hbar^2}{2mR^2}\frac{\partial^2}{\partial \phi^2}$, where R is the radius of the circle and ϕ is the angular coordinate. What boundary condition should the wavefunction $\psi(\phi)$ satisfy? Using the Schrödinger equation and the boundary condition, determine the wavefunctions and energies of the stationary states. Determine the degeneracy of each energy level.

Exercise 4.1.3: Consider the three-dimensional harmonic oscillator with potential $V(x, y, z) = \frac{1}{2}m\omega^2(x^2 + y^2 + z^2)$. Answer the following questions using cartesian coordinates, separation of variables, and your knowledge of the energies and wavefunctions of the one-dimensional simple harmonic oscillator. Write down the energy levels in terms of three quantum numbers n_1, n_2, and n_3. Write the same energy levels in terms of one quantum number n. Show that the nth energy level is $\frac{1}{2}(n+1)(n+2)$–fold degenerate.

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4.2 The Schrödinger equation in spherical coordinates

The Schrödinger equation in three dimensions in position representation in spherical coordinates is

$$i\hbar\frac{\partial\psi(\vec{r},t)}{\partial t} = -\frac{\hbar^2}{2m}\vec{\nabla}^2\psi(\vec{r},t) + V(\vec{r})\psi(\vec{r},t), \tag{4.2.1}$$

where $\vec{r} = (r,\theta,\phi)$, $\psi(\vec{r},t) = \psi(r,\theta,\phi,t)$ and

$$\vec{\nabla}^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2}{\partial\phi^2} \tag{4.2.2}$$

is the Laplacian operator.

For a time-independent and central potential, $V(\vec{r}) = V(r)$. In other words, the potential depends only of the radial coordinate r but not on θ and ϕ. The time-independent Schrödinger equation then becomes

$$-\frac{\hbar^2}{2mr^2}\left(\frac{\partial}{\partial r}\left(r^2\frac{\partial\psi(r,\theta,\phi)}{\partial r}\right) + \frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\psi(r,\theta,\phi)}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2\psi(r,\theta,\phi)}{\partial\phi^2}\right) + V(r)\psi(r,\theta,\phi) = E\psi(r,\theta,\phi). \tag{4.2.3}$$

Because the potential does not mix the different coordinates, we can make separation of variables and write

$$\psi(r,\theta,\phi) = R(r)Y(\theta,\phi). \tag{4.2.4}$$

The Schrödinger equation then reduces to

$$\frac{1}{R(r)}\left[\frac{d}{dr}\left(r^2\frac{dR(r)}{dr}\right) - \frac{2mr^2}{\hbar^2}(V(r)-E)\right]R(r) = -\frac{1}{Y(\theta,\phi)}\left[\frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{dY(\theta,\phi)}{d\theta}\right) + \frac{1}{\sin^2\theta}\frac{d^2Y(\theta,\phi)}{d\phi^2}\right]. \tag{4.2.5}$$

Each side must be a constant and equal. Thus, denoting the constant by $l(l+1)$ for convenience in notation later,

$$\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial Y(\theta,\phi)}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2Y(\theta,\phi)}{\partial\phi^2} = -l(l+1)Y(\theta,\phi). \tag{4.2.6}$$

and

$$\frac{d}{dr}\left(r^2\frac{dR(r)}{dr}\right) - \frac{2mr^2}{\hbar^2}(V(r)-E)R(r) = l(l+1)R(r). \tag{4.2.7}$$

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4.3 Orbital angular momentum

The orbital angular momentum in classical mechanics is defined by $\vec{L} = \vec{r} \times \vec{p}$. In quantum mechanics, this is promoted to an operator, and in position representation,

$$\hat{\vec{L}} = \hat{\vec{r}} \times \hat{\vec{p}}. \tag{4.3.1}$$

In cartesian coordinates, the components of the angular momentum operator are

$$\hat{L}_x = \hat{y}\hat{p}_z - \hat{z}\hat{p}_y, \quad \hat{L}_y = \hat{z}\hat{p}_x - \hat{x}\hat{p}_z, \quad \hat{L}_z = \hat{x}\hat{p}_y - \hat{y}\hat{p}_x, \tag{4.3.2}$$

where we need to distinguish the hat over an operator versus over a unit vector from context. Because the components of the position and the momentum operators do not commute, the components of the angular momentum operator do not commute either. In particular,

$$\begin{aligned} [\hat{L}_x, \hat{L}_y] &= [\hat{y}\hat{p}_z - \hat{z}\hat{p}_y, \hat{z}\hat{p}_x - \hat{x}\hat{p}_z] = \hat{y}[\hat{p}_z, \hat{z}]\hat{p}_x] + \hat{x}[\hat{z}, \hat{p}_z]\hat{p}_y \\ &= -i\hbar\hat{y}\hat{p}_x + i\hbar\hat{x}\hat{p}_y = i\hbar\hat{L}_z \end{aligned} \tag{4.3.3}$$

Similarly, and collecting,

$$[\hat{L}_x, \hat{L}_y] = i\hbar\hat{L}_z, \quad [\hat{L}_y, \hat{L}_z] = i\hbar\hat{L}_z, \quad [\hat{L}_z, \hat{L}_x] = i\hbar\hat{L}_y, \tag{4.3.4}$$

or

$$[\hat{L}_i, \hat{L}_j] = i\hbar\epsilon_{ijk}\hat{L}_k \tag{4.3.5}$$

with $i = 1, 2, 3$ representing the x, y, and z components respectively, and ϵ_{ijk} is the epsilon tensor with components defined by

$$\epsilon_{123} = \epsilon_{231} = \epsilon_{312} = 1, \tag{4.3.6}$$

it becomes -1 for odd permutation of indices, such as $\epsilon_{213} = -1$, and vanishes when any index is repeated, such as $\epsilon_{112} = 0$.

Furthermore, the square of the total angular momentum operator is

$$\hat{\vec{L}}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2, \tag{4.3.7}$$

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and it commutes with each component of the total angular momentum. For instance,

$$\begin{aligned}[\hat{\vec{L}}^2, \hat{L}_z] &= [\hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2, \hat{L}_z] = \hat{L}_x[\hat{L}_x, \hat{L}_z] + [\hat{L}_x, \hat{L}_z]\hat{L}_x + \hat{L}_y[\hat{L}_y, \hat{L}_z] + [\hat{L}_y, \hat{L}_z]\hat{L}_y \\ &= -i\hbar\hat{L}_x\hat{L}_y - i\hbar\hat{L}_y\hat{L}_x + i\hbar\hat{L}_y\hat{L}_x, + i\hbar\hat{L}_x\hat{L}_y = 0. \end{aligned} \tag{4.3.8}$$

Thus,

$$[\hat{\vec{L}}^2, \hat{L}_i] = 0. \tag{4.3.9}$$

Question 4.3.1: Out of the four operators, $\hat{\vec{L}}$, $\hat{L}_x$, $\hat{L}_y$, and $\hat{L}_z$, how many could we simultaneously diagonalize? (A). 0, (B). 1, (C). 2, (D). 2, (E). 4.

Question 4.3.2: Out of the four operators, $\hat{\vec{L}}^2$, $\hat{L}_x$, $\hat{L}_y$, and $\hat{L}_z$, how many could we simultaneously diagonalize? (A). 0, (B). 1, (C). 2, (D). 2, (E). 4.

In position representation,

$$\hat{\vec{L}} = \hat{\vec{r}} \times \hat{\vec{p}} = -i\hbar\vec{r} \times \vec{\nabla} \tag{4.3.10}$$

In spherical coordinates,

$$\begin{aligned}\hat{\vec{L}} &= -i\hbar r\hat{r} \times \left(\hat{r}\frac{\partial}{\partial r} + \hat{\theta}\frac{1}{r}\frac{\partial}{\partial\theta} + \hat{\phi}\frac{1}{r\sin\theta}\frac{\partial}{\partial\phi}\right) \\ &= -i\hbar\left(\hat{\phi}\frac{\partial}{\partial\theta} - \hat{\theta}\frac{1}{\sin\theta}\frac{\partial}{\partial\phi}\right). \end{aligned} \tag{4.3.11}$$

The components along cartesian unit vectors can be obtained using the unit vector relations,

$$\hat{r} = \sin\theta\cos\phi\,\hat{x} + \sin\theta\sin\phi\,\hat{y} + \cos\theta\,\hat{z} \tag{4.3.12}$$

$$\hat{\theta} = \cos\theta\cos\phi\,\hat{x} + \cos\theta\sin\phi\,\hat{y} - \sin\theta\,\hat{z} \tag{4.3.13}$$

$$\hat{\phi} = -\sin\phi\,\hat{x} + \cos\phi\,\hat{y}. \tag{4.3.14}$$

With these,

$$\begin{aligned}\hat{\vec{L}} &= -i\hbar\left((-\sin\phi\,\hat{x} + \cos\phi\,\hat{y})\frac{\partial}{\partial\theta} - (\cos\theta\cos\phi\,\hat{x} + \cos\theta\sin\phi\,\hat{y} - \sin\theta\,\hat{z})\frac{1}{\sin\theta}\frac{\partial}{\partial\phi}\right) \\ &= L_x\hat{x} + L_y\hat{y} + L_z\hat{z}, \end{aligned} \tag{4.3.15}$$

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where

$$\hat{L}_x = i\hbar\left(\sin\phi\,\frac{\partial}{\partial\theta} + \cot\theta\cos\phi\,\frac{\partial}{\partial\phi}\right) \tag{4.3.16}$$

$$\hat{L}_y = i\hbar\left(-\cos\phi\,\frac{\partial}{\partial\theta} + \cot\theta\sin\phi\,\frac{\partial}{\partial\phi}\right) \tag{4.3.17}$$

$$\hat{L}_z = -i\hbar\frac{\partial}{\partial\phi} \tag{4.3.18}$$

Squaring, $\hat{\vec{L}}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2$,

$$\hat{\vec{L}}^2 = -\hbar^2\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}\right] \tag{4.3.19}$$

With this, the angular part of the Schrödinger equation given by (4.2.6) can be rewritten as

$$\hat{\vec{L}}^2 Y(\theta,\phi) = l(l+1)\hbar^2 Y(\theta,\phi). \tag{4.3.20}$$

We also rearrange and rewrite the radial part given by (4.2.7) as

$$-\frac{\hbar^2}{2m}\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dR(r)}{dr}\right) + \left(V(r) + \frac{l(l+1)\hbar^2}{2mr^2}\right)R(r) = ER(r). \tag{4.3.21}$$

Or, rewriting again,

$$-\frac{\hbar^2}{2m}\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dR(r)}{dr}\right) - V_{\text{eff}}(r)R(r) = ER(r), \tag{4.3.22}$$

which is a Schrödinger like equation for the radial part of the wavefunction with effective potential,

$$V_{\text{eff}}(r) = V(r) + \frac{l(l+1)\hbar^2}{2mr^2}. \tag{4.3.23}$$

Note that the second term is the potential corresponding to centrifugal force in classical mechanics due to the orbital motion of the particle.

4.3.1 Spherical harmonics

We want to find the solutions to the angular equation $\hat{\vec{L}}^2 Y(\theta,\phi) = l(l+1)\hbar^2 Y(\theta,\phi)$; i.e.,

$$-\hbar^2\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}\right]Y(\theta,\phi) = l(l+1)\hbar^2 Y(\theta,\phi). \tag{4.3.24}$$

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Using separation of variables,

$$Y(\theta,\phi) = \Theta(\theta)\Phi(\phi) \tag{4.3.25}$$

we obtain

$$\frac{1}{\Theta}\Big[\sin\theta\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right) + l(l+1)\sin^2\theta\Big] = -\frac{1}{\Phi}\frac{d^2\Phi}{d\phi^2} = constant = m^2. \tag{4.3.26}$$

The solution for the ϕ equation has the simple form

$$\Phi(\phi) = e^{im\phi}. \tag{4.3.27}$$

Demanding that $\Phi(\phi)$ be periodic in ϕ with period 2π gives the constraint

$$m = 0, \pm1, \pm2, \cdots . \tag{4.3.28}$$

The equation for $\Theta(\theta)$ can be written in terms of a new variable defined by

$$x \equiv \cos\theta \tag{4.3.29}$$

as

$$(1-x^2)\frac{d^2P(x)}{dx^2} - 2x\frac{dP(x)}{dx} + \left(l(l+1) - \frac{m^2}{1-x^2}\right)P(x) = 0. \tag{4.3.30}$$

This is the associated Legendre differential equation whose solutions can be obtained using the method of power series. For the special case of $m = 0$, the equation is called the Legendre differential equation, and the solutions are called the Legendre polynomials.

Exercise 4.3.1: Solve the Legendre equation and construct the Legendre polynomials for the case of $m = 0$.

Solution: The Legendre equation is $(1-x^2)P'' - 2xP' + l(l+1)P = 0$. The idea is to use the ansatz $P(x) = \sum_{n=0}^{\infty} c_n x^n$ in the Legendre equation and organize it in the form $\sum_{n=0}^{\infty}[\ldots]_n x^n = 0$ which vanishes for all x, and thus solves the equation, when each coefficient $[\ldots]_n = 0$. Now $xP'(x) = \sum_{n=0}^{\infty} nc_n x^n$, $P'(x) = \sum_{n=0}^{\infty}(n-1)nc_n x^{n-2} = \sum_{n=2}^{\infty}(n-1)nc_n x^{n-2} = \sum_{n=0}^{\infty}(n+1)(n+2)c_{n+2}x^n$, and $x^2P'(x) = \sum_{n=0}^{\infty}(n-1)nc_n x^n$. Observe that we have rewritten the same expressions in order to produce the same

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power of x^n in each. With these, $P'' - x^2P'' - 2x\frac{dP}{dx} + l(l+1)P = 0$ becomes

$$\sum_{n=0}^{\infty} [(n+2)(n+1)c_{n+2} - n(n+1)c_n + l(l+1)c_n]\, x^n = 0, \tag{4.3.31}$$

which gives the recursion relation

$$\frac{c_{n+2}}{c_n} = \frac{n(n+1) - l(l+1)}{(n+2)(n+1)}. \tag{4.3.32}$$

The recursion relation gives the entire series in terms of two independent constants c_0 and c_1. Let us organize the even and odd terms separately:

$$P(x) = (c_0 + c_2x^2 + c_4x^4 + \cdots) + (c_1x + c_3x^3 + c_5x^5 + \cdots). \tag{4.3.33}$$

Noting that $\lim_{n\to\infty} \frac{c_{n+2}}{c_n} = 1$, the power series evidentally diverges at $x = \pm 1$, which is physically unacceptable in problems that include the z-axis. However, if l is an even integer and one chooses $c_1 = 0$, the series built on c_0 will be truncated after c_l, because the recursion relation implies c_{l+2} and all subsequent terms vanish. Likewise, if l is an odd integer and one chooses $c_0 = 0$, the odd series will be truncated after $n = l$, and the even series will vanish. In this way we obtain finite solutions labeled by the integer l, which are even or odd when l is even or odd. These are the Legendre polynomials $P_l(x)$, which are uniquely defined once we specify the normalization

$$\int_{-1}^{1} \mathrm{d}x\, P_l(x)P_{l'}(x) = \frac{2}{2l+1}\delta_{ll'}, \tag{4.3.34}$$

which fixes c_0 or c_1.

Truncating it at $n = 0$; i.e., $c_2 = 0$ while $c_0 \neq 0$, we get $l = 0$ fixed. Then, $P_0 = c_0$. Normalizing using the given convention, $\int_{-1}^{1} c_0^2(x)dx = c_0^2 2 = 2$, which gives $c_0 = 1$, where we choose the polynomials to be real functions. Therefore, $P_0 = 1$.

Truncating it at $n = 1$; i.e., $c_3 = 0$ while $c_1 \neq 0$, we get $l = 1$ fixed. Then, $P_1 = c_1x$. Normalizing, $\int_{-1}^{1} c_1^2x^2dx = c_1^2\frac{2}{3} = \frac{2}{3}$, which gives $c_1 = 1$. Therefore, $P_1 = x$.

Truncating it at $n = 2$; i.e., $c_4 = 0$ while $c_2 \neq 0$, we get $l = 2$ fixed. Then, $\frac{c_2}{c_0} = \frac{0(0+1)-2(2+1)}{(0+2)(0+1)} = -3$, or $c_2 = -3c_0$. Therefore, $P_3(x) = c_0(1 - 3x^2)$. Normalizing,

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$\int_{-1}^{1} c_0^2((1-3x^2)^2 dx = \frac{8}{5}c_0^2 = \frac{2}{5}$, which gives $c_0 = \pm\frac{1}{2}$ and we choose $c_0 = -\frac{1}{2}$ to get $P_2(x) = \frac{1}{2}(3x^2-1)$, etc.

These polynomials can be constructed using the Rodrigues formula,

$$P_l(x) = \frac{1}{2^l l!}(\frac{d}{dx})^l(x^2-1)^l. \tag{4.3.35}$$

In particular, $P_0(x) = \frac{1}{2^0 0!}(\frac{d}{dx})^0(x^2-1)^0 = 1$, $P_1(x) = \frac{1}{2^1 1!}(\frac{d}{dx})^1(x^2-1)^1 = \frac{1}{2}(\frac{d}{dx})(x^2-1) = x$, etc.

The solutions to the associated Legendre differential equation are called the associated Legendre functions, and with our phase convention,

$$\begin{aligned} P_l^{m_l}(x) &= \frac{1}{2^l l!}(1-x^2)^{\frac{|m_l|}{2}}(\frac{d}{dx})^{l+|m_l|}(x^2-1)^l \\ &= (1-x^2)^{\frac{|m_l|}{2}}(\frac{d}{dx})^{|m_l|}P_l(x). \end{aligned} \tag{4.3.36}$$

where $P_l(x)$ is the lth Legendre polynomial given by the Rodrigues formula

$$P_l(x) = \frac{1}{2^l l!}\left(\frac{d}{dx}\right)^l (x^2-1)^l \tag{4.3.37}$$

with l taking values

$$l = 0, 1, 2, ... \tag{4.3.38}$$

and for each l, m_l takes $2l+1$ values

$$m_l = -l, -l+1, ..., l-1, l. \tag{4.3.39}$$

The normalized solutions to the angular part, including both θ and ϕ, are the spherical harmonics,

$$Y_l^{m_l}(\theta,\phi) = (-1)^{m_l}\sqrt{\frac{(2l+1)}{4\pi}\frac{(l-m_l)!}{(l+m_l)!}}e^{im_l\phi}P_l^{m_l}(\cos\theta). \tag{4.3.40}$$

for $m_l \geq 0$, and

$$Y_l^{m_l}(\theta,\phi) = \sqrt{\frac{(2l+1)}{4\pi}\frac{(l-|m_l|)!}{(l+|m_l|)!}}e^{im_l\phi}P_l^{m_l}(\cos\theta). \tag{4.3.41}$$

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for $m_l < 0$. Alternatively, putting both negative and positive m_l together,

$$Y_l^{m_l}(\theta,\phi) = (-1)^{\frac{m_l+|m_l|}{2}} \sqrt{\frac{(2l+1)}{4\pi}\frac{(l-|m_l|)!}{(l+|m_l|)!}} e^{im_l\phi} P_l^{m_l}(\cos\theta). \tag{4.3.42}$$

$Y_l^{m_l}(\theta,\phi)$ are simultaneous eigenfunctions of $\hat{\vec{L}}^2$ and $\hat{L}_z$ with

$$\hat{\vec{L}}^2 Y_l^{m_l} = l(l+1)\hbar^2 Y_l^{m_l}, \qquad \hat{L}_z Y_l^{m_l} = m_l \hbar Y_l^{m_l}. \tag{4.3.43}$$

The first few Legendre polynomials are listed below.

$$\begin{aligned} P_0(x) &= 1 & P_0(\theta) &= 1 \\ P_1(x) &= x & P_1(\theta) &= \cos\theta \\ P_2(x) &= \frac{1}{2}(3x^2-1) & P_2(\theta) &= \frac{1}{2}(3\cos^2\theta - 1) \end{aligned} \tag{4.3.44}$$

The Legendre polynomials are normalized such that they satisfy the orthogonality relation

$$\int_{-1}^{1} P_{l'}(x) P_l(x)\, dx = \int_0^{\pi} P_{l'}(\theta) P_l(\theta) \sin\theta d\theta = \frac{2}{2l+1}\delta_{l'l} \tag{4.3.45}$$

The first few associated Legendre functions are listed below.

$$\begin{aligned} P_0^0(x) &= 1 & P_0^0(\theta) &= 1 \\ P_1^1(x) &= \sqrt{1-x^2} & P_1^1(\theta) &= \sin\theta \\ P_1^0(x) &= x & P_1^0(\theta) &= \cos\theta \\ P_2^2(x) &= 3(1-x^2) & P_2^2(\theta) &= 3\sin^2\theta \\ P_2^1(x) &= 3x\sqrt{1-x^2} & P_2^1(\theta) &= 3\cos\theta\sin\theta \\ P_2^0(x) &= \frac{1}{2}\left(3x^2-1\right) & P_2^0(\theta) &= \frac{1}{2}\left(3\cos^2\theta - 1\right) \end{aligned} \tag{4.3.46}$$

with

$$P_l^{-m_l}(x) = P_l^{m_l}(x). \tag{4.3.47}$$

The associated Legendre functions are normalized such that they satisfy the orthogonality re-

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lation

$$\int_{-1}^{1} P_{l'}^{m_l'}(x) P_l^{m_l}(x)\, dx = \int_0^{\pi} P_{l'}^{m_l'}(\theta) P_l^{m_l}(\theta) \sin\theta d\theta = \frac{(l+m_l)!}{(2l+1)(l-m_l)!} \delta_{l'l} \delta_{m_l' m_l} \tag{4.3.48}$$

The first few spherical harmonics are given by

$$\begin{aligned}
Y_0^0(\theta,\phi) &= \frac{1}{\sqrt{4\pi}} \\
Y_1^{\pm 1}(\theta,\phi) &= \mp \sqrt{\frac{3}{8\pi}} \sin\theta\, e^{\pm i\phi} \\
Y_1^0(\theta,\phi) &= \sqrt{\frac{3}{4\pi}} \cos\theta \\
Y_2^{\pm 2}(\theta,\phi) &= \sqrt{\frac{15}{32\pi}} \sin^2\theta\, e^{\pm 2i\phi} \qquad (4.3.49) \\
Y_2^{\pm 1}(\theta,\phi) &= \mp \sqrt{\frac{15}{8\pi}} \sin\theta\, e^{\pm i\phi} \qquad (4.3.50) \\
Y_2^0(\theta,\phi) &= \sqrt{\frac{5}{16\pi}} \left(3\cos^2\theta - 1\right) \qquad (4.3.51)
\end{aligned}$$

The spherical harmonics satisfy the orthogonality relation

$$\int_0^{2\pi} d\phi \int_0^{\pi} d\theta \sin\theta\, Y_{l'}^{m_l'*}(\theta,\phi) Y_l^{m_l}(\theta,\phi) = \delta_{l'l} \delta_{m_l' m_l}. \tag{4.3.52}$$

The eigenvalue problem of orbital angular momentum $\hat{\vec{L}}^2 Y_l^{m_l} = l(l+1)\hbar^2 Y_l^{m_l}$ and $\hat{L}_z Y_l^{m_l} = m\hbar Y_l^{m_l}$ was solved above in position representation. The quantum numbers, or eigenvalues for the simultaneously diagonalizable operators $\hat{\vec{L}}^2$ and $\hat{L}_z$, come in a pair of numbers l, m_l. We can write this eigenstate as a ket in abstract vector space as $|lm_l\rangle$, and we have

$$\hat{\vec{L}}^2 |lm_l\rangle = l(l+1)\hbar^2 |lm_l\rangle, \qquad \hat{L}_z |lm_l\rangle = m\hbar |lm_l\rangle. \tag{4.3.53}$$

The spherical harmonics are then the wavefunctions in position representation

$$Y_l^{m_l}(\theta,\phi) = \langle \vec{r} | lm_l \rangle. \tag{4.3.54}$$

Exercise 4.3.2:

Consider a "rigid rotor" model for the rotational motion of H_2 molecule with two

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H atoms, each of mass M, at fixed separation of $2R$.

1. Considering a molecule rotating about an axis perpendicular to the center of the interatomic separation, write down the classical Hamiltonian for the rotational energy of the rotor in terms of angular momentum $\vec{L}$ and moment of inertia $I = 2MR^2$.
2. Promote the classical Hamiltonian to a Hamiltonian operator, and write down the wavefunctions in spherical coordinates and the corresponding energies.
3. Using the mass of the proton for the mass of the hydrogen atom and taking $2R = 0.15\,nm$, determine the numerical value of the first four energy levels in eV.

4.3.2 Parity of the spherical harmonics

The parity operator takes $\vec{r}$ to $-\vec{r}$. In cartesian coordinates

$$\hat{P}\psi(x, y, z) = \psi(-x, -y, -z). \tag{4.3.55}$$

In spherical coordinates,

$$\hat{P}\psi(r, \theta, \phi) = \psi(r, \pi - \theta, \phi + \pi). \tag{4.3.56}$$

In particular, for the Legendre polynomials, noting that

$$P_l^{m_l}(\theta) = (1 - \cos^2\theta)^{|m_l|/2}\left(\frac{d}{d\cos\theta}\right)^{|m_l|}\frac{1}{2^l l!}\left(\frac{d}{d\cos\theta}\right)^l(\cos^2\theta - 1)^l \tag{4.3.57}$$

and $\hat{P}\cos\theta = \cos(\pi - \theta) = -\cos\theta$, we have

$$\hat{P}P_l^{m_l}(\theta) = P_l^{m_l}(\pi - \theta) = (-1)^{l+|m_l|}P_l^{m_l}(\theta) = (-1)^{l-|m_l|}P_l^{m_l}(\theta) \tag{4.3.58}$$

Therefore, the Legendre polynomials are even for $l + |m_l|$ even and odd for $l + |m_l|$ odd. Furthermore, for the azimuthal part of the wavefunction,

$$\hat{P}e^{im_l\phi} = e^{im_l(\phi+\pi)} = (-1)^{m_l}e^{im_l\phi}. \tag{4.3.59}$$

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The spherical harmonics are products of the two, and

$$\hat{P}Y_l^{m_l}(\theta,\phi) = Y_l^{m_l}(\pi-\theta,\phi+\pi) = (-1)^{l-|m_l|+m_l}Y_l^{m_l}(\theta,\phi) = (-1)^l Y_l^{m_l}(\theta,\phi). \tag{4.3.60}$$

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